

FRANSTA, Sweden

WMO: 023420

Lat: 62.517N

Lon: 16.183E

Elev: 105

StdP: 100.07

Time Zone: 1.00 (EUC)

Period: 82-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.8	-26.3	-31.6	0.2	-29.7	-28.2	0.3	-26.2	10.2	1.5	8.8	-2.3	0.0	20	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	26.3	17.0	24.1	16.1	21.9	14.9	18.3	24.3	16.9	22.2	15.7	20.3	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	11.4	20.2	14.6	10.5	18.9	13.5	9.8	17.8	52.1	24.3	47.9	22.4	44.2	20.4	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.4	7.1	5.4	-32.7	29.5	4.5	1.8	-35.9	30.8	-38.6	31.8	-41.1	32.9	-44.4	34.2
			-32.8	19.9	4.5	1.6	-36.0	21.0	-38.6	22.0	-41.1	22.9	-44.3	24.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-10.4	-8.0	-2.3	2.1	8.3	12.6	15.1	13.1	8.1	3.1	-3.2	-8.1
	DBStd	10.00	8.66	7.55	4.56	3.59	3.69	3.46	2.87	2.90	3.00	4.11	5.46	7.18
	HDD10.0	3099	634	503	382	237	79	12	1	5	73	215	396	562
	HDD18.3	5758	892	737	641	486	310	174	107	165	307	473	646	821
	CDD10.0	394	0	0	0	1	27	91	159	101	16	1	0	0
	CDD18.3	13	0	0	0	0	0	3	7	2	0	0	0	0
	CDH23.3	272	0	0	0	0	22	72	144	34	1	0	0	0
	CDH26.7	50	0	0	0	0	1	17	26	6	0	0	0	0
Wind	WSAvg	1.6	1.2	1.2	1.8	1.9	2.1	2.1	2.0	1.7	1.7	1.7	1.2	1.2
Precipitation	PrecAvg	596	37	24	25	33	42	65	92	85	51	55	45	40
	PrecMax	877	72	57	56	92	98	122	239	150	139	151	102	73
	PrecMin	444	5	2	4	3	13	20	25	35	16	14	12	4
	PrecStd	103	18	14	12	22	21	26	53	37	28	33	24	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	7.3	9.7	19.2	25.9	29.1	29.1	27.4	20.1	14.3	9.6	5.2
		MCWB	3.7	4.1	5.5	11.2	14.2	19.3	18.8	18.2	13.8	11.0	6.9	3.2
	2%	DB	4.0	5.0	7.0	14.1	22.9	25.2	26.8	23.5	17.3	12.2	6.7	3.7
		MCWB	1.8	2.4	3.3	8.0	13.5	16.8	17.1	16.6	12.5	9.2	5.1	2.2
	5%	DB	1.9	3.3	5.4	11.0	19.8	22.4	24.7	20.9	15.2	10.7	5.0	2.3
		MCWB	0.3	1.5	2.4	5.8	12.4	14.9	16.4	14.9	11.5	8.6	3.7	1.1
	10%	DB	0.4	1.6	3.8	8.8	17.1	20.2	22.2	19.1	13.4	9.3	3.5	0.9
		MCWB	-0.6	0.3	1.3	4.7	10.5	13.7	15.4	14.1	10.1	7.6	2.4	0.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	4.6	5.7	11.3	16.0	19.5	19.7	19.1	14.9	11.7	7.1	3.5
		MCDB	6.2	6.9	9.0	19.0	23.3	27.2	26.7	25.2	18.2	13.1	8.7	4.6
	2%	WB	2.1	2.9	4.1	8.4	14.2	17.8	18.5	17.4	13.5	10.2	5.2	2.4
		MCDB	3.6	4.5	6.3	13.3	21.2	24.2	24.6	21.8	16.3	11.4	6.6	3.4
	5%	WB	0.6	1.6	2.7	6.4	12.7	15.7	17.2	16.1	12.0	8.9	3.8	1.3
		MCDB	1.6	3.0	4.8	10.1	19.4	21.0	22.8	19.8	14.4	10.2	4.6	2.1
	10%	WB	-0.5	0.4	1.6	5.1	10.9	14.4	16.0	14.8	10.8	7.6	2.5	0.1
		MCDB	0.4	1.5	3.3	7.9	16.0	19.3	21.1	17.7	12.7	9.3	3.4	0.8
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.9	9.3	10.0	12.5	10.9	11.1	9.8	9.2	7.6	6.3	7.6
		MCDBR	8.4	8.8	10.9	15.3	17.9	15.6	16.4	13.3	12.1	9.8	6.5	7.4
		MCWBR	7.3	6.9	7.5	9.2	9.1	7.9	7.7	6.5	7.2	6.8	5.4	6.4
	5% WB	MCDBR	8.0	8.0	9.4	13.6	16.7	14.4	14.0	11.7	9.8	7.8	6.2	7.3
		MCWBR	7.2	6.7	6.8	8.8	8.6	8.0	7.5	6.4	6.3	6.1	5.5	6.4
Clear-Sky Solar Irradiance	taub	0.230	0.287	0.313	0.350	0.354	0.349	0.365	0.356	0.323	0.308	0.247	0.347	
	taud	2.090	2.233	2.311	2.371	2.426	2.470	2.443	2.461	2.542	2.455	2.177	2.652	
	Ebn at Noon	439	665	794	836	864	875	848	824	792	653	416	273	
	Edh at Noon	32	62	83	97	100	97	98	88	67	51	29	20	
All-Sky Solar Radiation	RadAvg	0.20	0.84	2.30	3.79	4.74	5.47	4.99	3.79	2.38	1.04	0.30	0.11	
	RadStd	0.02	0.13	0.24	0.32	0.49	0.37	0.56	0.32	0.29	0.18	0.06	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page